The Astrophysical Journal, 962:73 (9pp), 2024 February 10 https://doi.org/10.3847/1538-4357/ad1b4f
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Revised Constraints on the Fast Radio Burst Population from the First CHIME/FRB Catalog

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Received 2023 September 7; revised 2023 December 14; accepted 2024 January 3; published 2024 February 8

Abstract

In this paper, we investigate the fast radio burst (FRB) population using the first CHIME/FRB catalog. We first reconstruct the extragalactic dispersion measure–redshift relation ($\mathrm{DM_E}$–z relation) from well-localized FRBs, then use it to infer the redshift and isotropic energy of the first CHIME/FRB catalog. The intrinsic energy distribution is modeled by the power law with an exponential cutoff, and the selection effect of the CHIME telescope is modeled by a two-parametric function of specific fluence. For the intrinsic redshift distribution, the star formation history (SFH) model and five other SFH-related models are considered. We construct the joint likelihood of fluence, energy, and redshift, and all the free parameters are constrained simultaneously using the Bayesian inference method. The Bayesian information criterion (BIC) is used to choose the model that best matches the observational data. For comparison, we fit our models with two data samples, i.e., the Full sample and the Gold sample. The power-law index and cutoff energy are tightly constrained to be $1.8 \lesssim \alpha \lesssim 1.9$ and $\log(E_c/\mathrm{erg}) \approx 42$, which are almost independent of the redshift distribution model and the data sample we choose. The parameters involving the selection effect strongly depend on the data sample but are insensitive to the redshift distribution model. According to BIC, the pure SFH model is strongly disfavored by both the Full sample and Gold sample. For the other five SFH-related redshift distribution models, most of them can match the data well if the parameters are properly chosen. Therefore, with the present data, it is still premature to draw a conclusion on the FRB population.

Unified Astronomy Thesaurus concepts: Radio transient sources (2008); Intergalactic medium (813)

1. Introduction

Fast radio bursts (FRBs) are very bright radio pulses with a duration of milliseconds that randomly happen in the sky (Cordes & Chatterjee 2019; Petroff et al. 2019; Zhang 2020; Xiao et al. 2021). The first FRB was declared in 2007 when Lorimer et al. (2007) reexamined the 2001 archive data of the Parkes 64 m telescope in Australia. Until now, hundreds of FRBs have been detected by different telescopes (Petroff et al. 2016; Amiri et al. 2021), such as the Canadian Hydrogen Intensity Mapping Experiment (CHIME; Amiri et al. 2018) and the Five-hundred-meter Aperture Spherical Telescope (Nan et al. 2011), etc. Their large values of dispersion measure (DM) in excess of the contribution of the Milky Way indicate an extragalactic or even cosmological origin (Keane et al. 2016; Chatterjee et al. 2017; Tendulkar et al. 2017), except one that is verified to be generated from the Milky Way (Andersen et al. 2020; Bochenek et al. 2020). FRBs are generally classified into repeating and nonrepeating. Most of the sources observed by the CHIME telescope are apparently nonrepeating. The first repeating FRB, 121102, was discovered by the Arecibo radio telescope (Scholz et al. 2016; Spitler et al. 2016), and soon its host galaxy was identified and the cosmological origin was confirmed (Chatterjee et al. 2017; Marcote et al. 2017; Tendulkar et al. 2017). Since then, our knowledge of FRBs has significantly advanced.

A mystery that still puzzles scientists is the redshift distribution of FRBs, which is highly correlated with their progenitors; i.e., different origins produce different redshift distributions (Zhang et al. 2021a). For instance, if FRBs originate from magnetars, it is reasonable to assume that they follow the cosmic star formation history (SFH; Zhang et al. 2021a; Amiri et al. 2021; James et al. 2022) because galactic magnetars are believed to come from massive star deaths (Kaspi & Beloborodov 2017). Recently, the landmark identification of Galactic FRB 20200428A, together with an X-ray burst from the magnetar SGR 1935+2154, proved the magnetar origin of some, if not all, FRBs (Andersen et al. 2020; Bochenek et al. 2020; Mereghetti et al. 2020; Li et al. 2021; Ridnaia et al. 2021; Tavani et al. 2021). The hypothesis that most FRBs track SFH was further confirmed by the Parkes and ASKAP FRB samples (Zhang et al. 2021a; James et al. 2022). However, the SFH model fails to explain the large sample of CHIME/FRBs (Qiang et al. 2022; Zhang & Zhang 2022). The discovery of the repeating source FRB 20200120E situated in a globular cluster of the nearby galaxy M81 hints that some FRBs may track old stellar populations (Bhardwaj et al. 2021; Kirsten et al. 2022; Nimmo et al. 2022). In addition, delayed models that FRBs track the compact binary mergers are also acceptable (Cao et al. 2018; Locatelli et al. 2019; Zhang et al. 2021a; Zhang & Zhang 2022). Above all, the intrinsic redshift distribution of FRBs is still in extensive debate.

Though the number of detected FRBs is rapidly increasing due to the development of observation techniques, the sample that has direct redshift measurement is still small (Heintz et al. 2020). Several works use DM as a rough proxy of redshift, i.e., $\mathrm{DM} \approx 1000z\ \mathrm{pc\ cm^{-3}}$ (Lorimer 2018), as confirmed by theoretical analyses and multiple observations (Ioka 2003; Inoue 2004; Deng & Zhang 2014; Macquart et al. 2020). However, due to the complex components of DM, this formula

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is very rough and may cause significant bias if it is used to estimate redshift. For example, although the DM of FRB 20190520B is as large as 1200 pc cm^{-3}, the redshift is measured to be only $z \sim 0.24$ (Niu et al. 2022). A more reasonable way is to reconstruct the DM–z relation (the so-called Macquart relation) by properly taking the probability distributions of the DMs of the intergalactic medium (IGM) and host galaxy into account and to use the Macquart relation to infer redshift (Macquart et al. 2020).

Except for the redshift distribution, the intrinsic energy distribution is crucial for studying the FRB population as well. The intrinsic energy distribution is well constrained regardless of the redshift distribution. It is commonly believed to follow the power law $dN/dE \propto E^{-\alpha}$, with $1.8 \lesssim \alpha \lesssim 2.0$ (Luo et al. 2018, 2020, 2020; Lu & Piro 2019). Luo et al. (2020) and Lu et al. (2020) argued that there may be a high-energy exponential cutoff E_c, which is estimated to be around $\log(E_c/\mathrm{erg}) \approx 41.5$. The cutoff energy is still not tightly constrained, and other values are also acceptable (Zhang et al. 2021a). Note that the specific fluence is a convolution of isotropic energy and redshift. It should follow a simple law $N \propto F_\nu^{-3/2}$ and is insensitive to either redshift distribution or energy distribution (Zhang & Zhang 2022). The recently published CHIME/FRB catalog provides a relatively large sample for one to study the FRB population (Amiri et al. 2021). After considering the selection effect of the observational instrument, a properly chosen redshift distribution model should fit the data well. The aim of this paper is to choose a redshift distribution model that can best match the observational data.

In this paper, we first use the reconstructed DM$_E$–z relation to infer the redshift of CHIME/FRBs. Then the SFH model, along with five SFH-related models proposed by Qiang et al. (2022), is adopted to describe the possible redshift distribution. After considering the selection effect of the telescope, we construct the distributions of fluence, energy, and redshift. Finally, we use the Bayesian inference method to constrain the model parameters by fitting the models to the CHIME/FRBs. The rest of this paper is organized as follows. In Section 2, we reconstruct the DM$_E$–z relation from well-localized FRBs and then use it to infer the redshift and calculate the energy of the CHIME/FRBs. In Section 3, we consider six redshift distribution models and construct a Bayesian framework to constrain the parameters. Finally, discussion and conclusions are drawn in Section 4.

2. Inferring the Redshift of CHIME/FRB

The speed of electromagnetic waves traveling through cosmic plasma is frequency-dependent; that is, high-frequency electromagnetic waves travel faster than low-frequency ones. The relative delay of arrival time is proportional to the DM, which is defined as the integral of free electron number density along the line of sight. The observed DM of an extragalactic FRB can be decomposed into four terms (Deng & Zhang 2014; Gao et al. 2014; Macquart et al. 2020),

$$\mathrm{DM_{obs}} = \mathrm{DM_{MW,ISM}} + \mathrm{DM_{MW,halo}} + \mathrm{DM_{IGM}} + \frac{\mathrm{DM_{host}}}{1+z}, \tag{1}$$

where the subscripts "MW, ISM," "MW, halo," "IGM," and "host" denote contributions from the Milky Way interstellar medium (ISM), the Galactic halo, the IGM, and the FRB host galaxy, respectively. Notice that the DM$_{\mathrm{host}}$ is in the source frame, which is weighted by $(1+z)^{-1}$ to transform it to the observer frame, with z being the source redshift.

The Milky Way ISM term (DM$_{\mathrm{MW,ISM}}$) can be estimated using Milky Way electron density models, such as the NE2001 model (Cordes & Lazio 2002) and the YMW16 model (Yao et al. 2017). These two models predict consistent results for FRBs at high Galactic latitude, but the YMW16 model may overestimate the DM$_{\mathrm{MW,ISM}}$ value for FRBs at low Galactic latitude (Koch Ocker et al. 2021). Therefore, we use the NE2001 model to calculate DM$_{\mathrm{MW,ISM}}$ in this work. The Galactic halo term is estimated to be about DM$_{\mathrm{MW,halo}} \approx$ 50 pc cm^{-3}, although the exact value of this term is still poorly constrained (Prochaska & Zheng 2019; Macquart et al. 2020). Here we follow Macquart et al. (2020) to fix DM$_{\mathrm{MW,halo}} =$ 50 pc cm^{-3}. We emphasize that the concrete value of DM$_{\mathrm{MW,halo}}$ should not strongly bias our analyses, since its uncertainty is much smaller than that of the last two terms, i.e., DM$_{\mathrm{IGM}}$ and DM$_{\mathrm{host}}$, which will be discussed in detail below.

By subtracting the first two terms from the total DM$_{\mathrm{obs}}$, we define the extragalactic DM as

$$\begin{aligned}\mathrm{DM}_E &\equiv \mathrm{DM_{obs}} - \mathrm{DM_{MW,ISM}} - \mathrm{DM_{MW,halo}} \\ &= \mathrm{DM_{IGM}} + \frac{\mathrm{DM_{host}}}{1+z}.\end{aligned} \tag{2}$$

Only the term DM$_{\mathrm{IGM}}$ contains cosmological information, the expression of which in the flat ΛCDM model reads (Deng & Zhang 2014; Zhang et al. 2021a)

$$\langle \mathrm{DM_{IGM}}(z) \rangle = \frac{3cH_0\Omega_b f_{\mathrm{IGM}} f_e}{8\pi G m_p} \int_0^z \frac{1+z}{\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}} dz, \tag{3}$$

with c the speed of light, m_p the proton mass, and G the Newtonian gravitational constant. $f_e = Y_{\mathrm{H}} X_{e,\mathrm{H}}(z) + \frac{1}{2} Y_{\mathrm{He}} X_{e,\mathrm{He}}(z)$ is the electron fraction, denoting the extent of ionization progress of hydrogen and helium. Y_{H} and Y_{He} are the cosmological mass fractions of hydrogen and helium, which are estimated to be around ∼0.75 and ∼0.25, respectively. Given that the electrons in both hydrogen and helium are fully ionized at $z < 3$ (Meiksin 2009; Becker et al. 2011), we take the ionization fractions to be $X_{e,\mathrm{H}} = X_{e,\mathrm{He}} = 1$. The Hubble constant H_0, the baryon mass density Ω_b, the matter density Ω_m, and the vacuum energy density Ω_Λ are all fixed to the Planck 2018 parameters, i.e., $H_0 = 67.4$ km s^{-1} Mpc^{-1}, $\Omega_m =$ 0.315, $\Omega_\Lambda = 0.685$, and $\Omega_b = 0.0493$ (Aghanim et al. 2020). As for the cosmic baryon mass fraction f_{IGM}, we know little about the evolution model of this term at present. Several observations indicate that f_{IGM} is about 0.84 (Fukugita et al. 1998; Inoue 2004). With five well-localized FRBs detected by ASKAP, Li et al. (2020) also drew a similar conclusion. Therefore, we fix $f_{\mathrm{IGM}} = 0.84$ throughout this paper.

Though the average value of DM$_{\mathrm{IGM}}$ is given by Equation (3), the real value of it should oscillate around the mean owing to the large-scale matter density fluctuation. Theoretical analyses as well as numerical simulations show that the distribution of DM$_{\mathrm{IGM}}$ can be modeled as a long-tailed quasi-Gaussian distribution (McQuinn 2014; Macquart et al.

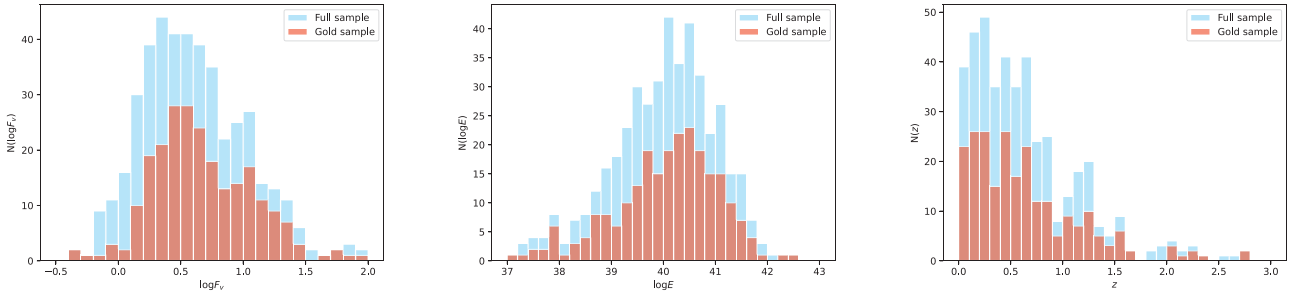

Figure 1. The histograms of fluence ($\log F_\nu$), isotropic energy ($\log E$), and inferred redshift (z) of the first nonrepeating CHIME/FRB catalog.

2020; Zhang et al. 2021b),

$$p_{\rm IGM}(\Delta) = A\Delta^{-\beta}\exp\left[-\frac{(\Delta^{-\alpha}-C_0)^2}{2\alpha^2\sigma_{\rm IGM}^2}\right],\quad \Delta>0, \tag{4}$$

where $\Delta \equiv {\rm DM_{IGM}}/\langle {\rm DM_{IGM}}\rangle$, and $\sigma_{\rm IGM} = Fz^{-0.5}$ denotes the effective standard deviation, with F a free parameter that quantifies the strength of baryon feedback (Macquart et al. 2020). α and β are two parameters related to the inner density profile of gas in halos, which are chosen to be $\alpha=\beta=3$ (Macquart et al. 2020). C_0 and A are constants dependent on $\sigma_{\rm IGM}$, where C_0 is chosen to guarantee that the mean value of Δ is unity and A to normalize the distribution accordingly.

At present, there is still little knowledge of the host term ${\rm DM_{host}}$ due to the absence of detailed observation of the local environment of most FRB sources. It may vary on a large scale, ranging from several tens (Xu et al. 2022) to several hundreds (Niu et al. 2022) of pc cm^{-3}. To account for the large variation of ${\rm DM_{host}}$, it is usually modeled by the lognormal distribution (Macquart et al. 2020; Zhang et al. 2020),

$$p_{\rm host}({\rm DM_{host}}|\mu,\sigma_{\rm host}) = \frac{1}{\sqrt{2\pi}\,{\rm DM_{host}}\sigma_{\rm host}}\exp\left[-\frac{(\ln{\rm DM_{host}}-\mu)^2}{2\sigma_{\rm host}^2}\right], \tag{5}$$

with μ and $\sigma_{\rm host}$ the mean and standard deviation of $\ln{\rm DM_{host}}$, respectively. At present, there is no strong evidence for the redshift dependence of μ or $\sigma_{\rm host}$ (Tang et al. 2023). Thus, we treat these two parameters as constants.

After properly taking the distributions of ${\rm DM_{IGM}}$ and ${\rm DM_{host}}$ into account, one can construct the distribution of ${\rm DM}_E$ at redshift z by marginalizing over either term as (Macquart et al. 2020)

$$p_E({\rm DM}_E|z) = \int_0^{(1+z){\rm DM}_E} p_{\rm host}({\rm DM_{host}}|\mu,\sigma_{\rm host})p_{\rm IGM}\left({\rm DM}_E - \frac{{\rm DM_{host}}}{1+z}|F,z\right)d{\rm DM_{host}}, \tag{6}$$

where $(F,\mu,\sigma_{\rm host})$ are free parameters that can be constrained by fitting to a sample of well-localized FRBs.

Using 17 well-localized FRBs, Tang et al. (2023) get the parameters $(F,\mu,\sigma_{\rm host})$ well constrained. By fixing the parameters to the best-fitting values, Equation (6) can be used to reconstruct the ${\rm DM}_E$–z relation; see Figure 2 of Tang et al. (2023). Then the ${\rm DM}_E$–z relation can be used to infer the redshift of FRBs that have no direct measurement of redshift but have well-determined ${\rm DM}_E$, e.g., the first CHIME/FRB catalog.

The first CHIME/FRB catalog consists of 536 bursts, with 474 apparently nonrepeating bursts and 62 repeating bursts from 18 FRB sources (Amiri et al. 2021). All of the bursts have well-detected ${\rm DM_{obs}}$, but most of them are without a redshift measurement. We calculate the ${\rm DM}_E$ of the 474 apparently nonrepeating FRBs by subtracting ${\rm DM_{MW,ISM}}$ and ${\rm DM_{MW,halo}}$ from the total ${\rm DM_{obs}}$. Then the ${\rm DM}_E$–z relation is used to calculated the redshift of FRBs with ${\rm DM}_E > 100$ pc cm^{-3} (444 FRBs in total). In terms of FRBs with ${\rm DM}_E < 100$ pc cm^{-3}, the ${\rm DM_{host}}$ term dominates over ${\rm DM_{IGM}}$, which may cause a large uncertainty on the redshift estimation. The inferred redshifts are listed in the online materials of Tang et al. (2023). The inferred redshifts are in the range of $z_{\rm inf} \in (0.023, 3.935)$. Only three bursts have inferred redshifts larger than 3, i.e., FRB 20180906B with $z_{\rm inf} = 3.935^{+0.463}_{-0.705}$, FRB 20181203C with $z_{\rm inf} = 3.003^{+0.443}_{-0.657}$, and FRB 20190430B with $z_{\rm inf} = 3.278^{+0.449}_{-0.650}$. For the sake of statistical significance, we omit the three sparse data points at redshift $z>3$ in the following FRB population analyses.

With the observed fluence listed in the CHIME/FRB catalog and the inferred redshift, the isotropic energy of an FRB can be calculated as (James et al. 2021)

$$E = \frac{4\pi d_{\rm L}^2}{(1+z)^{2+\beta}}F_\nu\Delta\nu, \tag{7}$$

where $d_{\rm L}$ is the luminosity distance, β is the spectrum index ($F_\nu \propto \nu^\beta$), and $\Delta\nu$ is the bandwidth in which the FRB is detected. Macquart et al. (2019) showed that $\beta=-1.5$ performs a good fit based on a sample of ASKAP/FRBs; hence, we fix $\beta=-1.5$ in this work. In the first CHIME/FRB catalog, the observed fluence is averaged in the range $400\sim800$ MHz, indicating $\Delta\nu = 400$ MHz. The 436 FRBs with fluence, energy, and redshift are called the Full sample and are collected in the online material of Tang et al. (2023).

Furthermore, we apply the following criteria put forward by Amiri et al. (2021) to filter the data, in order to avoid bias in FRB population analyses: (1) ${\rm S/N} > 12$, (2) ${\rm DM_{obs}} > 1.5\max({\rm DM_{NE2001}}, {\rm DM_{YMW16}})$, (3) not detected in far sidelobes, (4) not detected during nonnominal telescope operations, and (5) $\tau_{\rm scat} < 10$ ms. The 236 FRBs filtered by the criteria are called the Gold sample, which will be used below to study the FRB population and confront the results of using the Full sample. We uniformly divide the redshift z into 30 bins with step size $\Delta z = 0.1$ in the range (0, 3), divide the energy $\log E$ into 30 bins with step size $\Delta\log E = 0.2$ in the range (37, 43), and divide the fluence $\log F_\nu$ into 25 bins with step size

$\Delta \log F_\nu = 0.1$ in the range $(-0.5, 2)$. The histograms of these properties of the Full and Gold samples are shown in Figure 1.

3. The FRB Population

As now we have the distributions of fluence, energy, and redshift, we can use them to constrain the FRB population. Several independent works regardless of the redshift distribution show that the intrinsic energy distribution roughly follows a power law $dN/dE \propto E^{-\alpha}$, with $1.8 \lesssim \alpha \lesssim 2.0$ (Luo et al. 2018, 2020; Lu & Piro 2019; Lu et al. 2020). It may have a high-energy exponential cutoff E_c (Lu et al. 2020; Luo et al. 2020), which is not yet well constrained (Zhang et al. 2021a). Therefore, we model the intrinsic energy distribution as the cutoff power law,

$$p(E) \propto \left(\frac{E}{E_c}\right)^{-\alpha} \exp\left(-\frac{E}{E_c}\right). \tag{8}$$

The redshift distribution in the observer frame can be derived from the intrinsic event-rate density $dN/dtdV$ according to the relation (Zhang et al. 2021a; Zhang & Zhang 2022)

$$p(z) \propto \frac{dt}{dt_{\rm obs}} \frac{dN}{dtdV} \frac{dV}{dz}, \tag{9}$$

where $dt/dt_{\rm obs} = (1+z)^{-1}$ accounts for the cosmic expansion, and the redshift-dependent specific comoving volume dV/dz in the flat ΛCDM model can be expressed as

$$\frac{dV}{dz} = \frac{4\pi c d_{\rm C}^2}{H_0 \sqrt{\Omega_m (1+z)^3 + \Omega_\Lambda}}, \tag{10}$$

where $d_{\rm C}$ is the comoving distance.

Actually, the intrinsic event-rate density $dN/dtdV$ is poorly constrained. The most well-motivated model is the SFH model. Several forms regarding the SFH have been proposed; e.g., Yuksel et al. (2008) fits a wide range of SFH data with a three-segment empirical model, and a two-segment empirical model proposed in Madau & Dickinson (2014) is also widely adopted, an updated version of which is used in Madau & Fragos (2017). However, no proof shows that there is significant priority among these forms. For consistency, we follow Qiang et al. (2022) to adopt the updated version of the two-segment empirical SFH model, i.e.,

$$\mathrm{SFH} = \frac{(1+z)^{2.6}}{1 + ((1+z)/3.2)^{6.2}}. \tag{11}$$

Several studies show that the intrinsic event-rate density does not simply track SFH (Qiang et al. 2022; Zhang & Zhang 2022). It is argued in Kulkarni (2018) that the research history of FRBs may resemble that of GRBs, which is found to follow an enhanced evolution with regard to SFH, i.e., $dN/dtdV \propto (1+z)^{1.5} \times \mathrm{SFH}(z)$ (Kistler et al. 2008; Yuksel et al. 2008). This prompts us to take some similar SFH-related models into consideration. In this work, we reconsider the revised SFH-related redshift distribution models put forward by Qiang et al. (2022). The models are enumerated below.

1. Power-law (PL) model:

$$\frac{dN}{dtdV} \propto (1+z)^{\gamma} \times \mathrm{SFH}(z). \tag{12}$$

2. Power law with an exponential cutoff (CPL) model:

$$\frac{dN}{dtdV} \propto (1+z)^{\gamma} \exp\left(-\frac{z}{z_c}\right) \times \mathrm{SFH}(z). \tag{13}$$

3. SFH with an exponential cutoff (CSFH) model:

$$\frac{dN}{dtdV} \propto \exp\left(-\frac{z}{z_c}\right) \times \mathrm{SFH}(z). \tag{14}$$

4. Two-segment (TSE) model:

$$\frac{dN}{dtdV} \propto \frac{(1+z)^{\gamma_1}}{1 + ((1+z)/s)^{\gamma_1+\gamma_2}} \times \mathrm{SFH}(z). \tag{15}$$

5. Non-SFH-based two-segment redshift distribution (TSRD) model:

$$\frac{dN}{dtdV} \propto \frac{(1+z)^{a}}{1 + ((1+z)/C)^{a+b}}. \tag{16}$$

We should note that Equations (8) and (9) do not actually provide the energy and redshift distributions detected by the telescope. To transform them to the actually detected distributions, the selection effect should be considered. Unfortunately, the selection effect of the CHIME telescope is hard to determine. For simplicity, we take the selection model proposed by Zhang & Zhang (2022). This model assumes the so-called "gray zone" between the minimum specific threshold fluence, i.e., $\log F_{\nu,\rm th}^{\rm min} = -0.5$ (as shown in the CHIME catalog), and the maximum specific threshold fluence $\log F_{\nu,\rm th}^{\rm max}$. In the "gray zone," FRBs are not fully detected. The detection efficiency is modeled by the power-law form,

$$\eta_{\rm det}(F_\nu) = \left(\frac{\log F_\nu - \log F_{\nu,\rm th}^{\rm min}}{\log F_{\nu,\rm th}^{\rm max} - \log F_{\nu,\rm th}^{\rm min}}\right)^n, \quad \log F_{\nu,\rm th}^{\rm min} < \log F_\nu < \log F_{\nu,\rm th}^{\rm max}. \tag{17}$$

For $\log F_\nu \leqslant \log F_{\nu,\rm th}^{\rm min}$, FRBs are undetectable, hence $\eta_{\rm det} = 0$; while for $\log F_\nu \geqslant \log F_{\nu,\rm th}^{\rm max}$, FRBs are fully detected, hence $\eta_{\rm det} = 1$. In Zhang & Zhang (2022), $n = 3$ is fixed, and the parameter $\log F_{\nu,\rm th}^{\rm max}$ is manually adjusted to fit the observed fluence distribution. Here we treat both n and $\log F_{\nu,\rm th}^{\rm max}$ as free parameters. The specific fluence F_ν can be written as a function of isotropic energy and redshift according to Equation (7), which reads

$$F_\nu = \frac{(1+z)^{2+\beta}}{4\pi d_{\rm L}^2 \Delta\nu} E. \tag{18}$$

Since the redshift distribution and intrinsic energy distribution are independent of each other, the joint distribution of redshift and energy can be simply written as $p(z, E) = p(z)p(E)$. Then, according to the probability theory, the detected fluence distribution after considering the selection effect can be written as

$$p_{\rm det}(F_\nu) = \frac{d}{dF_\nu} \iint_{F_\nu(E,z) \leqslant F_\nu} p(z)p(E)\eta_{\rm det}(F_\nu(E,z))\, dE\, dz, \tag{19}$$

where the two-dimensional integral region is defined by $F_\nu(E, z) \leqslant F_\nu$, with $F_\nu(E, z)$ given by Equation (18). Furthermore, the

Table 1
The Best-fitting Parameters and Their 1σ Confidence Levels by Simultaneously Constraining Three Sets of Parameters Using the Gold Sample

Model (1)	α (2)	$\log E_c/\mathrm{erg}$ (3)	$\log F_{\nu,\mathrm{th}}^{\max}$ (4)	n (5)	Model Parameter (6)	(7)	(8)	ΔBIC (9)
SFH	$2.13^{+0.01}_{-0.01}$	$42.03^{+0.01}_{-0.01}$	$0.66^{+0.01}_{-0.01}$	$4.45^{+0.05}_{-0.05}$	N/A	N/A	N/A	555.15
PL	$1.83^{+0.01}_{-0.01}$	$42.07^{+0.01}_{-0.01}$	$0.46^{+0.01}_{-0.01}$	$5.27^{+0.19}_{-0.23}$	$\gamma = -2.16^{+0.06}_{-0.06}$	N/A	N/A	34.25
CSFH	$1.87^{+0.01}_{-0.01}$	$42.15^{+0.02}_{-0.02}$	$0.47^{+0.02}_{-0.01}$	$5.45^{+0.28}_{-0.33}$	$z_c = 0.93^{+0.03}_{-0.03}$	N/A	N/A	0 (fiducial)
CPL	$1.87^{+0.01}_{-0.01}$	$42.15^{+0.02}_{-0.02}$	$0.47^{+0.02}_{-0.01}$	$5.43^{+0.31}_{-0.33}$	$\gamma = -0.04^{+0.30}_{-0.28}$	$z_c = 0.94^{+0.15}_{-0.11}$	N/A	6.57
TSE	$1.89^{+0.01}_{-0.01}$	$42.13^{+0.02}_{-0.02}$	$0.49^{+0.01}_{-0.02}$	$5.24^{+0.37}_{-0.27}$	$\gamma_1 = 7.14^{+1.95}_{-2.15}$	$\gamma_2 = 2.25^{+0.11}_{-0.09}$	$s = 1.05^{+0.02}_{-0.02}$	2.25
TSRD	$1.88^{+0.01}_{-0.01}$	$42.15^{+0.02}_{-0.02}$	$0.48^{+0.02}_{-0.01}$	$5.44^{+0.29}_{-0.31}$	$a = 3.28^{+0.82}_{-0.65}$	$b = 1.35^{+0.39}_{-0.33}$	$C = 1.50^{+0.19}_{-0.14}$	7.91

Note. Column (1): model name. Columns (2) and (3): parameters in energy distribution. Columns (4) and (5): parameters in selection function. Columns (6)–(8): parameters in different redshift distribution models. Column (9): ΔBIC with CSFH the fiducial model.

Table 2
The Best-fitting Parameters and Their 1σ Confidence Levels by Fitting to the Full Sample

Model	α	$\log E_c/\mathrm{erg}$	$\log F_{\nu,\mathrm{th}}^{\max}$	n	Model Parameter			ΔBIC
SFH	$2.14^{+0.01}_{-0.01}$	$41.96^{+0.01}_{-0.01}$	$0.70^{+0.01}_{-0.01}$	$3.53^{+0.02}_{-0.02}$	N/A	N/A	N/A	1997.01
PL	$1.78^{+0.01}_{-0.01}$	$42.12^{+0.01}_{-0.01}$	$0.42^{+0.01}_{-0.01}$	$3.79^{+0.07}_{-0.07}$	$\gamma = -2.53^{+0.03}_{-0.03}$	N/A	N/A	75.91
CSFH	$1.84^{+0.01}_{-0.01}$	$42.04^{+0.01}_{-0.01}$	$0.49^{+0.01}_{-0.01}$	$3.46^{+0.07}_{-0.05}$	$z_c = 0.84^{+0.02}_{-0.01}$	N/A	N/A	39.38
CPL	$1.81^{+0.01}_{-0.01}$	$42.08^{+0.01}_{-0.01}$	$0.46^{+0.01}_{-0.01}$	$3.57^{+0.06}_{-0.07}$	$\gamma = -0.99^{+0.15}_{-0.16}$	$z_c = 1.35^{+0.16}_{-0.12}$	N/A	25.12
TSE	$1.83^{+0.01}_{-0.01}$	$42.05^{+0.01}_{-0.01}$	$0.48^{+0.01}_{-0.01}$	$3.50^{+0.10}_{-0.08}$	$\gamma_1 = 9.13^{+0.65}_{-1.15}$	$\gamma_2 = 2.49^{+0.05}_{-0.05}$	$s = 1.03^{+0.01}_{-0.01}$	0 (fiducial)
TSRD	$1.82^{+0.01}_{-0.01}$	$42.07^{+0.01}_{-0.01}$	$0.47^{+0.01}_{-0.01}$	$3.58^{+0.08}_{-0.07}$	$a = 3.06^{+0.54}_{-0.53}$	$b = 1.39^{+0.22}_{-0.17}$	$C = 1.39^{+0.12}_{-0.08}$	2.54

Note. Given in the same manner as Table 1, but ΔBIC is given with TSE the fiducial model.

detected energy distribution can be written as

$$p_{\mathrm{det}}(E) = \int_{z_{\min}}^{z_{\max}} p(z)p(E)\eta_{\mathrm{det}}(F_\nu(E, z))\, dz. \tag{20}$$

Similarly, the detected redshift distribution reads

$$p_{\mathrm{det}}(z) = \int_{E_{\min}}^{E_{\max}} p(z)p(E)\eta_{\mathrm{det}}(F_\nu(E, z))\, dE. \tag{21}$$

We can further construct the cumulative distribution function (CDF) of each quantity,

$$N_{\mathrm{det}}(>Q) = A\int_{Q}^{Q_{\max}} p_{\mathrm{det}}(Q)dQ, \quad Q = \{F_\nu, E, z\}, \tag{22}$$

where Q is the quality that represents F_ν, E, or z, and A is the normalization factor scaled to the total number of data points.

Equation (22) can be used to fit the observed distribution of fluence, energy, and redshift of the CHIME/FRBs. In order to avoid the arbitrariness of the binning, the CDF rather than the differential distribution is used in the fitting. The χ^2 of each distribution can be written as

$$\chi_Q^2 = \sum_i^n \frac{[N_{\mathrm{det}}(>Q_i) - N(>Q_i)]^2}{\sigma_i^2}, \quad Q = \{F_\nu, E, z\}, \tag{23}$$

where N is the number of FRBs whose F_ν, E, or z is larger than that of the ith FRB. The uncertainty is given by $\sigma = \sqrt{N}$ (Wang & Zhang 2019). The total χ^2 is constructed as

$$\chi_{\mathrm{total}}^2 = \chi_{F_\nu}^2 + \chi_z^2 + \chi_E^2, \tag{24}$$

based on which the likelihood of observing a sample of FRBs with F_ν, E, and z can be written as

$$\mathcal{L}(\mathrm{FRBs}|\mathrm{parameters}) \propto \exp\left(-\frac{1}{2}\chi_{\mathrm{total}}^2(F_\nu, E, z)\right), \tag{25}$$

where the parameters are the power-law index α and cutoff energy E_c for energy distribution, n and $\log F_{\nu,th}^{\max}$ for selection function, and the parameters in different redshift distribution models listed in Equations (12)–(16). Then the posterior probability distribution can be constructed according to the Bayesian theorem,

$$P(\mathrm{parameters}|\mathrm{FRBs}) \propto \mathcal{L}(\mathrm{FRBs}|\mathrm{parameters})P_0(\mathrm{parameters}), \tag{26}$$

where P_0 is the prior of the parameters.

Through Equation (26), one can calculate the best-fitting parameters and their 1σ confidence level with the public Python package emcee (Foreman-Mackey et al. 2013). We take flat priors for all the parameters in a broad range, i.e., $\alpha \in U(1, 5)$, $\log E_c \in U(39, 44)$, $n \in U(1, 8)$, $\log F_{\nu,th}^{\max} \in (-0.49, 2.0)$, γ, γ_1, γ_2, a, $b \in U(-10, 10)$, and z_c, s, $C \in U(0.1, 10)$. The posterior probability distribution functions (PDFs) of the model parameters and the two-dimensional confidence contours are shown in Figures 3–5 in the Appendix. The median values and 1σ uncertainties of the parameters are summarized in Tables 1 and 2 for the Gold sample and the Full sample, respectively. The best-fitting lines are depicted in Figure 2.

To quantify the goodness of fit and choose the best model, we calculate the Bayesian information criterion (BIC), which is defined by Schwarz (1978),

$$\mathrm{BIC} = -2\ln\mathcal{L}_{\max} + k\ln(N), \tag{27}$$

where $\mathcal{L}_{\max}$ is the maximum likelihood calculated at the best-fitting parameters, k is the number of free parameters, and N is the number of data points. The model with a smaller value of BIC reflects a better fit. Note that it is not the absolute but the relative value of BIC that matters. Given the fiducial model, the

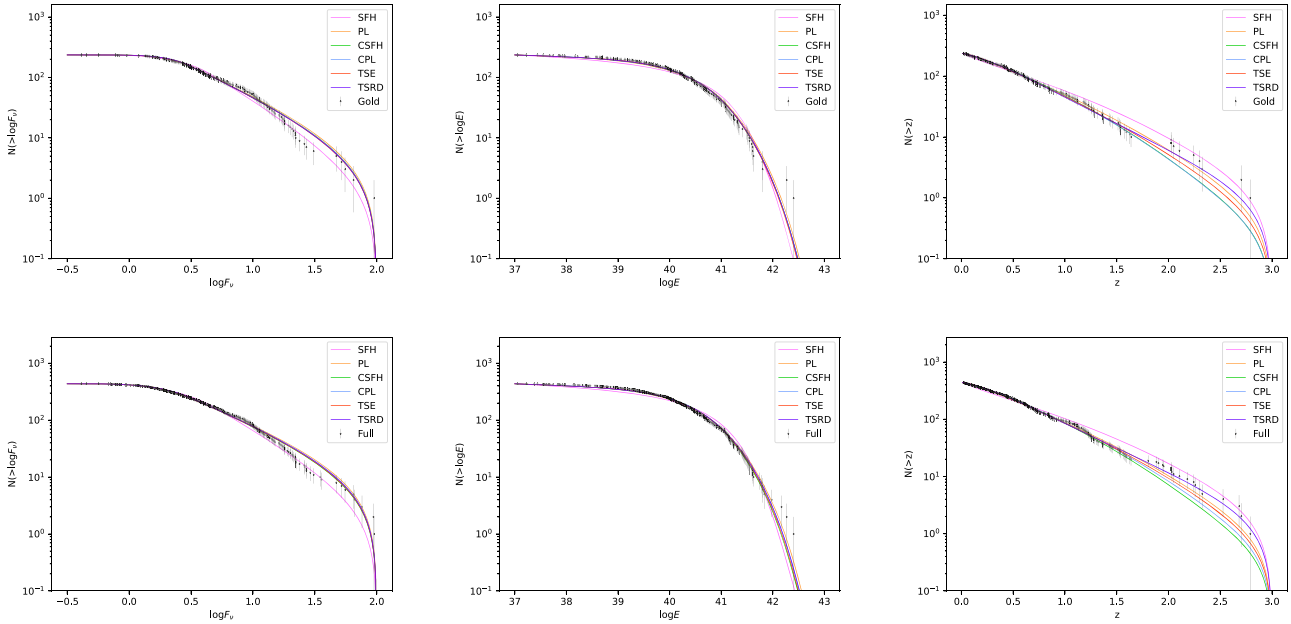

Figure 2. The best-fitting lines of the Gold sample (top panels) and the Full sample (bottom panels). The lines are plotted using Equation (22) with the best-fitting parameters given in Tables 1 and 2. Figures from left to right show the fits to fluence, energy, and redshift, respectively.

Table 3
The $p_{\rm ks}$ Values for $\log F_\nu$, $\log E$, and z of Different Redshift Models for the Gold Sample

Model	$p_{\rm ks}$ for $\log F_\nu$	$p_{\rm ks}$ for $\log E$	$p_{\rm ks}$ for z
SFH	0.2009	0.6014	0.3561
PL	0.8270	0.2345	0.9124
CSFH	0.7108	0.3755	0.9706
CPL	0.5685	0.3911	0.9648
TSE	0.6566	0.4647	0.6509
TSRD	0.6250	0.3800	0.8563

Table 4
The $p_{\rm ks}$ Values for $\log F_\nu$, $\log E$, and z of Different Redshift Models for the Full Sample

Model	$p_{\rm ks}$ for $\log F_\nu$	$p_{\rm ks}$ for $\log E$	$p_{\rm ks}$ for z
SFH	0.1879	3.58×10^{-8}	0.0189
PL	0.5886	0.0190	0.6780
CSFH	0.6781	0.0239	0.7282
CPL	0.7542	0.0309	0.9011
TSE	0.7027	0.0614	0.5358
TSRD	0.7656	0.0341	0.6445

relative value of BIC is defined as

$$\Delta{\rm BIC}_{\rm model} = {\rm BIC}_{\rm model} - {\rm BIC}_{\rm fiducial}. \quad (28)$$

According to the Jeffreys scale (Jeffreys 1998; Liddle 2007), $\Delta{\rm BIC} \geqslant 5$ or 10 indicates that there is "strong" or "decisive" evidence against the considered model with respect to the fiducial model. For the Gold sample, we take CSFH as the fiducial model, for it has the smallest BIC. For the Full sample, we take TSE as the fiducial model for the same reason. We list the ΔBIC values for each model in the last column of Tables 1 and 2.

With the results listed in Tables 1 and 2, we have the rank of fits CSFH $\approx$ TSE $>$ CPL $\approx$ TSRD $>$ PL $\gg$ SFH for the Gold sample and TSE $\approx$ TSRD $>$ CPL $>$ CSFH $>$ PL $\gg$ SFH for the Full sample. According to the ΔBIC values, both samples strongly disfavor the SFH model, compared with the other five models. However, a distinct difference exists between the Gold sample and Full sample. For example, the CSFH model is the best model for the Gold sample, while it is the third-worst model for the Full sample. In addition, the selection function parameter n differs significantly between the Gold sample and Full sample.

Furthermore, we perform the Kolmogorov–Smirnov (K-S) test in the same manner as in Qiang et al. (2022) to check if the best-fitting models can pass the statistical test or not. To this end, we confront the data points with the best-fitting CDF curve of each model and calculate the p-values ($p_{\rm ks}$), which are summarized in Table 3 for the Gold sample and Table 4 for the Full sample. The null hypothesis that the data points are drawn from the model distribution cannot be rejected if $p_{\rm ks} >$ 0.1 or 0.05. As can be seen, for the Gold sample, all models (even the worst model SFH) successfully pass the K-S test, although some models have smaller $p_{\rm ks}$ values than that of Qiang et al. (2022). However, the situation is different for the Full sample. All models except for SFH successfully pass the K-S test for fluence and redshift distributions, but none of the models can pass the K-S test for energy distribution. This is because in the Bayesian framework, the parameters are optimized by minimizing the total $\chi^2_{\rm total}$ (Equation (24)), rather than the individual $\chi^2(F_\nu)$, $\chi^2(E)$, or $\chi^2(z)$, which may induce bias for some observables. In this case, the K-S test may be conflicted with the results inferred from the Bayesian method.

4. Discussion and Conclusions

In this paper, we investigate the FRB population based on the first CHIME/FRB catalog using the Bayesian inference method. We first reconstruct the DM$_E$–z relation from 17 well-localized FRBs, then use it to infer the redshift of the CHIME/FRBs. Together with the observed fluence, the isotropic energy of each FRB is calculated. Thus, each FRB contains three observables: the redshift z, the fluence F_ν, and the isotropic

energy E. Finally, we construct a Bayesian framework to constrain the FRB population using the first CHIME/FRB catalog. We apply a set of criteria to exclude FRBs that may bias the population analysis and call the remaining FRBs the Gold sample. Three factors can affect the observables of FRBs: the intrinsic energy distribution, the selection effect of the detector, and the intrinsic redshift distribution. The intrinsic energy distribution is modeled by a power law with an exponential cutoff. The selection effect is modeled as a two-parametric function of fluence. The intrinsic redshift distribution is the factor that we are most interested in. The SFH model and five SFH-related redshift distribution models proposed by Qiang et al. (2022) are considered here. We construct a joint likelihood of fluence, energy, and redshift. All the free parameters are constrained simultaneously using the Bayesian inference method. The best-fitting parameters are summarized in Tables 1 and 2 for the Gold sample and Full sample, respectively.

From Tables 1 and 2, one can see that the energy distribution parameters (α and $\log E_c$) are tightly constrained. These two parameters are very stable; that is, they are almost independent of the models and data samples. Except for the SFH model, which has a power-law index $\alpha \approx 2.1$, the indices of the other five models fall into the range $1.8 \lesssim \alpha \lesssim 1.9$, well consistent with the previous findings (Luo et al. 2018, 2020; Lu & Piro 2019; Lu et al. 2020). For the cutoff energy E_c, it is tightly constrained to be about $\log E_c \approx 42$ for both the Full sample and Gold sample, which is slightly larger than the value ($\log E_c \approx 41$ or 41.5) used in Zhang & Zhang (2022) and Qiang et al. (2022).

The selection function parameters ($\log F_{\nu,\rm th}^{\rm max}$ and n) are also tightly constrained. For the Gold sample, these two parameters are very stable, with $\log F_{\nu,\rm th}^{\rm max} \approx 0.47$ and $n \approx 5.2 \sim 5.5$. However, the SFH model has a larger value of $\log F_{\nu,\rm th}^{\rm max} \approx 0.66$ and a smaller value of $n \approx 4.45$. The value of n we obtained here is about twice that used by Zhang & Zhang (2022) and Qiang et al. (2022). This is because Zhang & Zhang (2022) and Qiang et al. (2022) fixed $n = 3$ based on the full sample. As we can see from Table 2, the n value constrained from the Full sample is in the range of $3.4 \lesssim n \lesssim 3.8$, consistent with the value used by Zhang & Zhang (2022) and Qiang et al. (2022). The best-fitting value of n differs significantly between the Gold sample and Full sample. As is shown in the first histogram of Figure 1, the Gold sample shows a steeper incline at low fluence and peaks slightly later than the Full sample, which can be properly translated to a larger n and a slightly larger average $\log F_{\nu,\rm th}^{\rm max}$, in accordance with the best-fitting results. Qiang et al. (2022) fixed $\log E_c = 41.5$ and obtained the maximum specific threshold fluence $\log F_{\nu,\rm th}^{\rm max} \sim 0.75$; here we get a smaller $\log F_{\nu,\rm th}^{\rm max}$ but a larger $\log E_c$. The reason is that a smaller $\log F_{\nu,\rm th}^{\rm max}$ leads to a larger average fluence, which also requires a larger E_c to match the distribution.

As for the redshift distribution models, the one-parametric PL model and CSFH model can be tightly constrained. However, some parameters in multiparametric models have large uncertainties, such as γ in the CPL model, γ_1 in the TSE model, and a in the TSRD model. It is good to mention that we get results quite different from those in Qiang et al. (2022), where these models are initially proposed. For the one-parametric PL model and CSFH model, we get $\gamma = -2.16^{+0.06}_{-0.06}$ and $z_c = 0.93^{+0.03}_{-0.03}$ from the Gold sample, while Qiang et al. (2022) prefer larger values of $\gamma \approx -1.1$ and $z_c \approx 2.8$. For the two-parametric CPL model, we have $\gamma = -0.04^{+0.30}_{-0.28}$ and $z_c = 0.94^{+0.15}_{-0.11}$ from the Gold sample, in contrast to $\gamma \approx -0.6$ and $z_c \approx 5.5$ in Qiang et al. (2022). For the three-parametric TSE model and TSRD model, the distinction between our work and Qiang et al. (2022) is even larger, though the uncertainty becomes larger accordingly.

Several reasons may play important roles in these differences. First, we infer redshifts of CHIME/FRBs with the distribution of $\rm DM_{IGM}$ and $\rm DM_{host}$ properly considered. In contrast, Qiang et al. (2022) assume that $\rm DM_{host}$ is a constant and $\rm DM_{IGM}$ is given by the mean value and then calculate redshift by numerically solving Equation (1). The method of Qiang et al. (2022) may cause significant bias on the redshift inference and may lead to negative redshift for FRBs with small observed DM. Note that our method may still introduce some uncertainties, especially at the high-redshift region. This is because the $\rm DM_E$–z relation derived from the low-redshift data may be broken at the high-redshift region. Nevertheless, there is no strong evidence for the redshift evolution of host DM within the present data (Tang et al. 2023). Therefore, the $\rm DM_E$–z relation is still the best way to estimate the redshift of unlocalized FRBs. With more localized FRBs in the future, this problem could be alleviated. Second, the 452 CHIME/FRBs they used include both repeaters and nonrepeaters. In contrast, we only consider the nonrepeaters with $\rm DM_E > 100\,pc\,cm^{-3}$. The repeaters and nonrepeaters may have different populations, and the redshift estimation of FRBs with small $\rm DM_E$ may be significantly biased. Third, Qiang et al. (2022) fixed E_c, α, and n to reduce freedom and generated a large number of mock FRBs at each grid in parameter space to confront the real sample by performing a K-S test, which not only consumes computational power but also reduces precision. In contrast, we construct the joint distribution of fluence, energy, and redshift analytically; thus, all the parameters can be constrained simultaneously using the Bayesian inference method. Last but not least, they take priority in fitting the $N \propto F_\nu^{-3/2}$ relation, but it might be more reasonable to equally weight the fluence, energy, and redshift (or DM) distribution.

We also perform a K-S test to check if the best-fitting models can match the data points or not. We find that all models can successfully pass the K-S test for the Gold sample. For the Full sample, however, none of the models can simultaneously pass the K-S test for fluence, energy, and redshift distributions. It is possible to adjust the model parameters such that all three observables can simultaneously pass the K-S test, but this compromises the status of these parameters as "best-fitting." The Bayesian method can provide us the "best-fitting" parameters but without the commitment that all three observables simultaneously pass the K-S test. On the contrary, the K-S test method adopted by Zhang & Zhang (2022) and Qiang et al. (2022) can manually adjust the parameters to ensure that all three observables simultaneously satisfy the K-S test but have difficulty in finding the "best-fitting" parameters. In addition, the K-S test cannot deal with the uncertainty of observations, and model selection based on a K-S test may lead to contradiction. For instance, based on the $p_{\rm ks}$ values of the Gold sample (Table 3), the PL model seems to fit the fluence distribution best, while the energy distribution prefers the SFH model, but the CSFH model matches the redshift distribution best. Therefore, we use BIC rather than the K-S test to judge the goodness of fit.

Above all, the nonparametric SFH model is strongly disfavored by both the Gold sample and Full sample, as is shown by the ΔBIC values in Tables 1 and 2. According to BIC, the CSFH model performs the best fit to the Gold sample, while the TSE model fits the Full sample best. However, other models can also fit the data well, as long as the selection function of the detector and parameters is properly chosen. For example, the TSE model and CSFH model match the Gold sample equally well, while the TSRD model and TSE model match the Full sample equally well; see the ΔBIC values in Tables 1 and 2. Therefore, we summarize that it is still premature to make a conclusive determination about the FRB population with the present data, but the hypothesis that FRBs follow the SFH can be definitely excluded. We hope that a large number of well-localized FRBs observed in the near future can unambiguously determine the FRB population.

Acknowledgments

This work has been supported by the National Natural Science Fund of China under grant Nos. 12147102, 12275034, and 12347101, and the Fundamental Research Funds for the Central Universities of China under grant No. 2023CDJXY-048.

Appendix
The Contour Plots

The contour plots for the SFH, PL, CSFH, CPL, TSE, and TSRD models are shown in Figures 3–5. For a specific model, the constraints from the Full sample and Gold sample are plotted together for comparison.

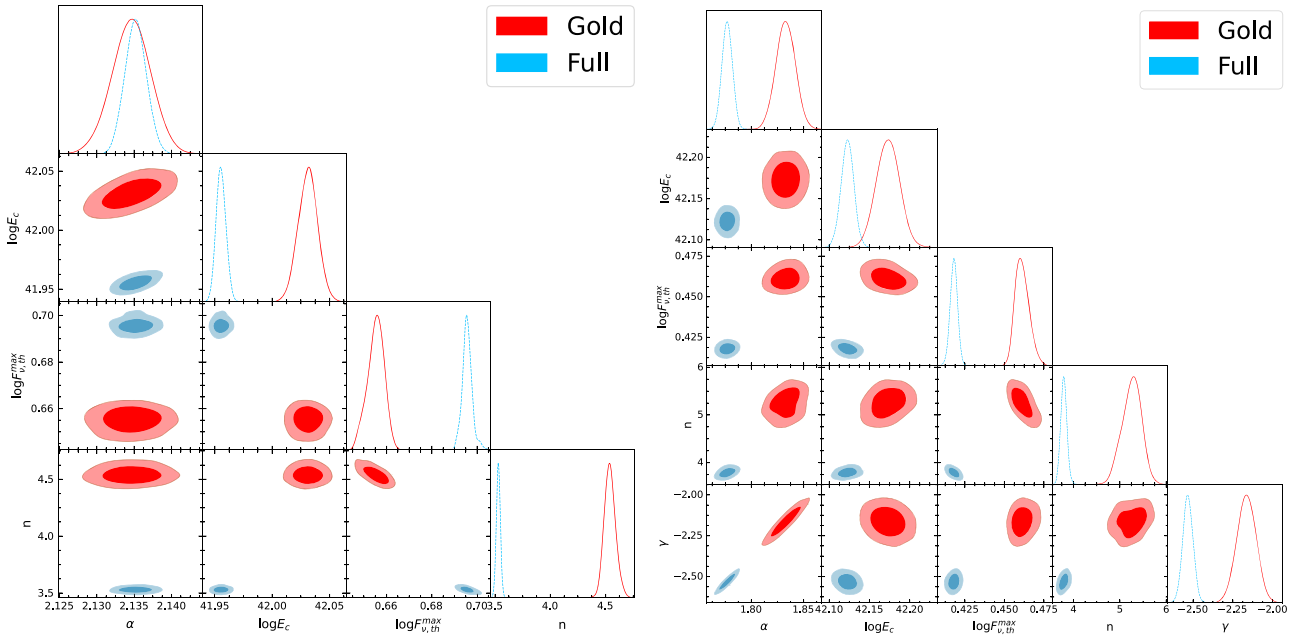

Figure 3. The contour plot for the SFH model (left) and PL model (right).

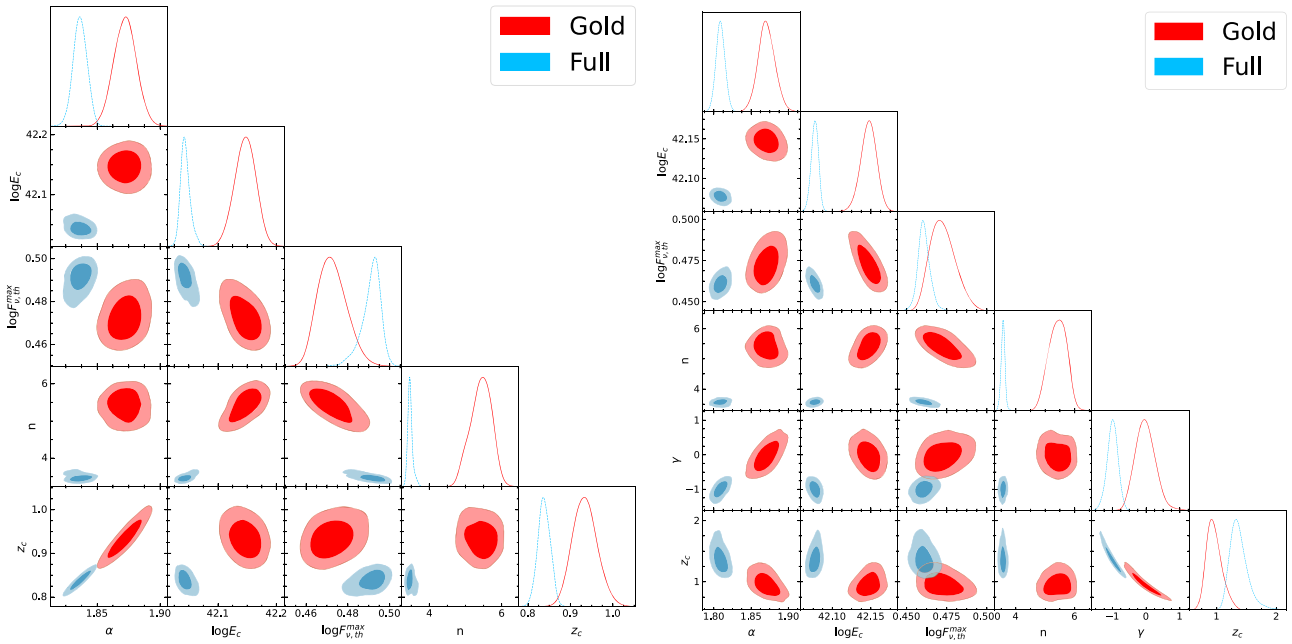

Figure 4. The contour plot for the CSFH model (left) and CPL model (right).

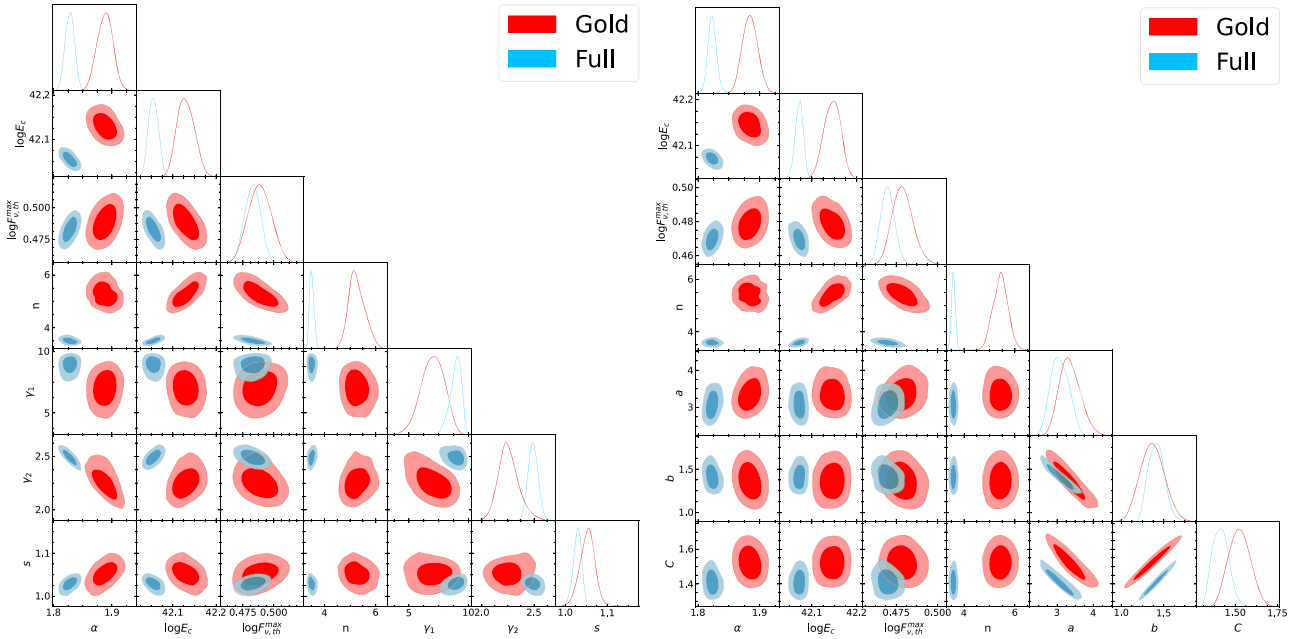

Figure 5. The contour plot for the TSE model (left) and TSRD model (right).

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References

Aghanim, N., Akrami, Y., Ashdown, M., et al. 2020, A&A, 641, A6
Amiri, M., Andersen, B. C., Bandura, K., et al. 2021, ApJS, 257, 59
Amiri, M., Bandura, K., Berger, P., et al. 2018, ApJ, 863, 48
Andersen, B. C., Bandura, K., Bhardwaj, M., et al. 2020, Natur, 587, 54
Becker, G. D., Bolton, J. S., Haehnelt, M. G., & Sargent, W. L. W. 2011, MNRAS, 410, 1096
Bhardwaj, M., Gaensler, B. M., Kaspi, V. M., et al. 2021, ApJL, 910, L18
Bochenek, C. D., Ravi, V., Belov, K. V., et al. 2020, Natur, 587, 59
Cao, X.-F., Yu, Y.-W., & Zhou, X. 2018, ApJ, 858, 89
Chatterjee, S., Law, C. J., Wharton, R. S., et al. 2017, Natur, 541, 58
Cordes, J. M., & Chatterjee, S. 2019, ARA&A, 57, 417
Cordes, J. M., & Lazio, T. J. W. 2002, arXiv:astro-ph/0207156
Deng, W., & Zhang, B. 2014, ApJL, 783, L35
Foreman-Mackey, D., Hogg, D. W., Lang, D., & Goodman, J. 2013, PASP, 125, 306
Fukugita, M., Hogan, C. J., & Peebles, P. J. E. 1998, ApJ, 503, 518
Gao, H., Li, Z., & Zhang, B. 2014, ApJ, 788, 189
Heintz, K. E., Prochaska, J. X., Simha, S., et al. 2020, ApJ, 903, 152
Inoue, S. 2004, MNRAS, 348, 999
Ioka, K. 2003, ApJL, 598, L79
James, C. W., Prochaska, J. X., Macquart, J. P., et al. 2021, MNRAS, 509, 4775
James, C. W., Prochaska, J. X., Macquart, J. P., et al. 2022, MNRAS, 510, L18
Jeffreys, H. 1998, The Theory of Probability (Oxford: Oxford Univ. Press)
Kaspi, V. M., & Beloborodov, A. 2017, ARA&A, 55, 261
Keane, E. F., Johnston, S., Bhandari, S., et al. 2016, Natur, 530, 453
Kirsten, F., Marcote, B., Nimmo, K., et al. 2022, Natur, 602, 585
Kistler, M. D., Yuksel, H., Beacom, J. F., & Stanek, K. Z. 2008, ApJL, 673, L119
Koch Ocker, S., Cordes, J. M., & Chatterjee, S. 2021, ApJ, 911, 102
Kulkarni, S. R. 2018, NatAs, 2, 832
Li, C. K., Lin, L., Xiong, S. L., et al. 2021, NatAs, 5, 378
Li, Z., Gao, H., Wei, J.-J., et al. 2020, MNRAS, 496, L28
Liddle, A. R. 2007, MNRAS: Lett., 377, L74
Locatelli, N., Ronchi, M., Ghirlanda, G., & Ghisellini, G. 2019, A&A, 625, A109
Lorimer, D. R. 2018, NatAs, 2, 860
Lorimer, D. R., Bailes, M., McLaughlin, M. A., Narkevic, D. J., & Crawford, F. 2007, Sci, 318, 777
Lu, W., Kumar, P., & Zhang, B. 2020, MNRAS, 498, 1397
Lu, W., & Piro, A. L. 2019, ApJ, 883, 40
Luo, R., Lee, K., Lorimer, D. R., & Zhang, B. 2018, MNRAS, 481, 2320
Luo, R., Men, Y., Lee, K., et al. 2020, MNRAS, 494, 665
Macquart, J. P., Prochaska, J. X., McQuinn, M., et al. 2020, Natur, 581, 391
Macquart, J. P., Shannon, R. M., Bannister, K. W., et al. 2019, ApJL, 872, L19
Madau, P., & Dickinson, M. 2014, ARA&A, 52, 415
Madau, P., & Fragos, T. 2017, ApJ, 840, 39
Marcote, B., Paragi, Z., Hessels, J. W. T., et al. 2017, ApJL, 834, L8
McQuinn, M. 2014, ApJL, 780, L33
Meiksin, A. A. 2009, RvMP, 81, 1405
Mereghetti, S., Savchenko, V., Ferrigno, C., et al. 2020, ApJL, 898, L29
Nan, R., Li, D., Jin, C., et al. 2011, IJMPD, 20, 989
Nimmo, K., Hessels, J. W. T., Kirsten, F., et al. 2022, NatAs, 6, 393
Niu, C. H., Aggarwal, K., Li, D., et al. 2022, Natur, 606, 873
Petroff, E., Barr, E. D., Jameson, A., et al. 2016, PASA, 33, e045
Petroff, E., Hessels, J. W. T., & Lorimer, D. R. 2019, A&ARv, 27, 4
Prochaska, J. X., & Zheng, Y. 2019, MNRAS, 485, 648
Qiang, D.-C., Li, S.-L., & Wei, H. 2022, JCAP, 01, 040
Ridnaia, A., Svinkin, D., Frederiks, D., et al. 2021, NatAs, 5, 372
Scholz, P., Spitler, L. G., Hessels, J. W. T., et al. 2016, ApJ, 833, 177
Schwarz, G. 1978, AnSta, 6, 461
Spitler, L. G., et al. 2016, Natur, 531, 202
Tang, L., Lin, H.-N., & Li, X. 2023, ChPhC, 47, 085105
Tavani, M., Casentini, C., Ursi, A., et al. 2021, NatAs, 5, 401
Tendulkar, S. P., Bassa, C. G., Cordes, J. M., et al. 2017, ApJL, 834, L7
Wang, F. Y., & Zhang, G. Q. 2019, ApJ, 882, 108
Xiao, D., Wang, F., & Dai, Z. 2021, SCPMA, 64, 249501
Xu, H., Niu, J. R., Chen, P., et al. 2022, Natur, 609, 685
Yao, J. M., Manchester, R. N., & Wang, N. 2017, ApJ, 835, 29
Yuksel, H., Kistler, M. D., Beacom, J. F., & Hopkins, A. M. 2008, ApJL, 683, L5
Zhang, B. 2020, Natur, 587, 45
Zhang, G. Q., Yu, H., He, J. H., & Wang, F. Y. 2020, ApJ, 900, 170
Zhang, R. C., & Zhang, B. 2022, ApJL, 924, L14
Zhang, R. C., Zhang, B., Li, Y., & Lorimer, D. R. 2021a, MNRAS, 501, 157
Zhang, Z. J., Yan, K., Li, C. M., Zhang, G. Q., & Wang, F. Y. 2021b, ApJ, 906, 49